

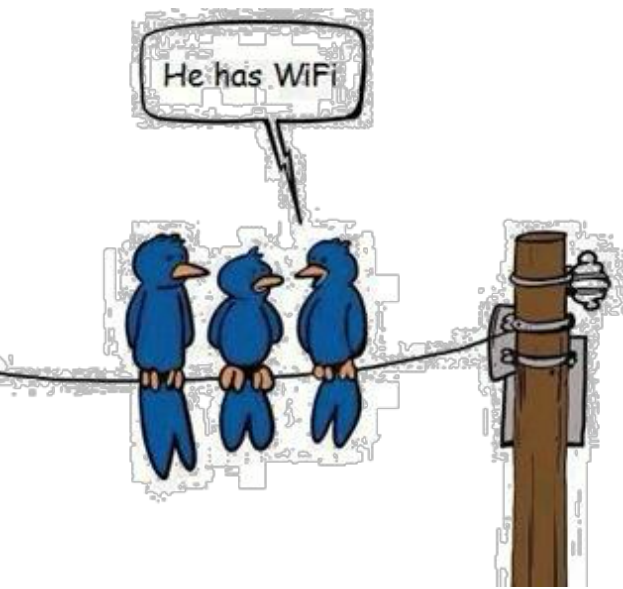
ECSE 4560/ 6560

Modern Communication Systems

Lecture 5: Wireless Systems Overview

Part 4 - Sampling

Ish Jain, Fall 2025



Imaging and Sensing for Next-generation Wireless Networks

Ashutosh Sabharwal

Ernest D. Butcher Professor in the Electrical and
Computer Engineering
Rice University

Mercer Distinguished Lecture Series

~~CII 3051~~ **SAGE 3510**

Wed, September 17, 2025 at 4:00 PM

As the wireless standards occupy larger parts of the spectrum with an ever-larger number of antennas, there is an opportunity to also use the communications spectrum for “imaging” the environment. Thus, there is an opportunity for next-generation networks like 6G to be multi-function, i.e., use the same spectrum for multiple functions. In this talk, we will show how multipath propagation can be harnessed to improve radar-based sensing to (i) image beyond the radar's field of view and (ii) estimate the full velocity vector using multipath reflections. We will also touch upon our recent work on converting base stations into weather radars.

Ashutosh Sabharwal is the Ernest D. Butcher Professor in the Electrical and Computer Engineering department at Rice University. He works in two independent areas - wireless networks and digital health. His ongoing work in wireless focuses on theory and methods for joint wireless communications & imaging, as well as large-scale experimental platforms. He was one of the inventors of in-band full-duplex communications, a technology used in wireline and wireless standards. He founded the WARP project (warp.rice.edu), an open-source project used by 150+ research groups worldwide. He also led Rice RENEW (renew-wireless.org), which developed an open-source software-defined massive MIMO wireless network platform. Concurrent with his wireless research, he also leads the Rice Digital Health Initiative (dhi.rice.edu) and is the Co-director of a new Methodist-Rice Digital Health Institute. His digital health research focus is the development of devices and data science to quantify behavior-biology pathways across many diseases. He received the 2017 IEEE Jack Neubauer Memorial Award, the 2018 IEEE Advances in Communications Award, the 2019, 2021, and 2025 ACM Sigmobile Test-of-time Awards, the 2019 ACM MobiCom Community Contribution Award, and the 2023 ICC Best Paper Award. His research has led to multiple startups and FDA-approved products. He is a Fellow of IEEE, ACM, and the National Academy of Inventors.



Announcements

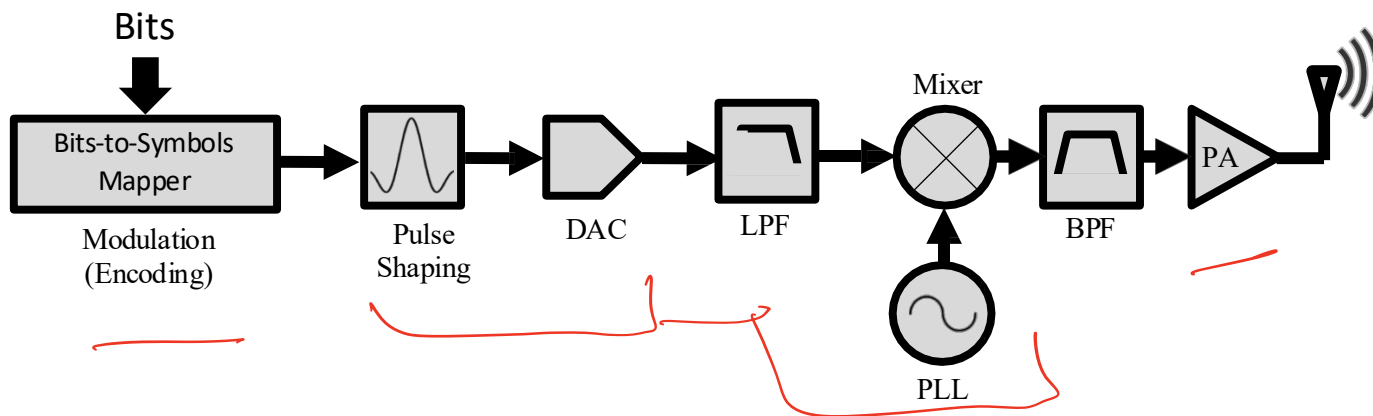
- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Sampling**
- Projects
 - A list of papers will be posted
 - Group formation [MCS Fall25 project.xlsx](#)
 - Proposal due 9/26
- HW1/Lab1 due Monday 9/22
 - Submit on Gradescope – Mark pages for each question parts
 - Matlab coding exercises – “Use Publish option to submit your code”

Today's Class: Wireless Overview 4

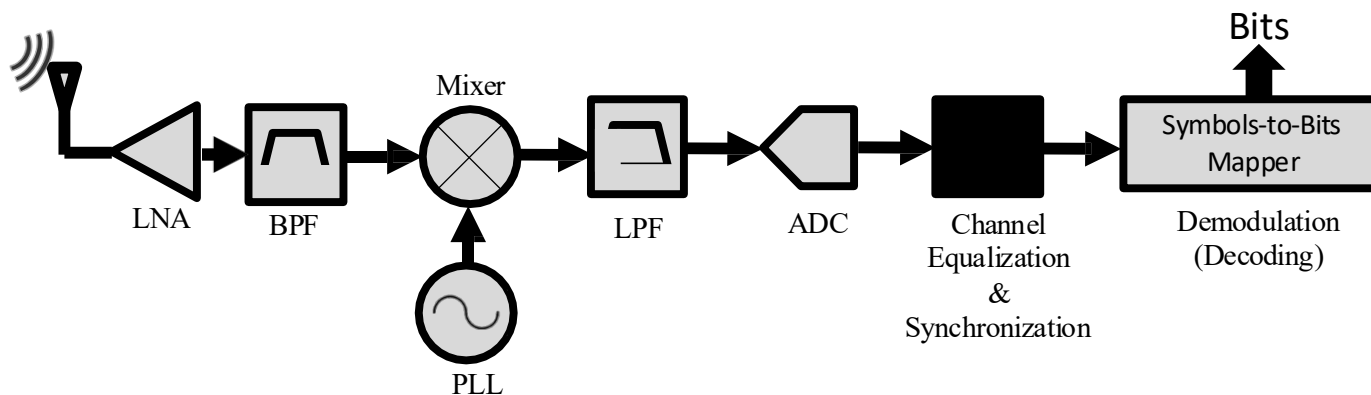
- Bit error rate and symbol error rate (Review)
- Review Fourier Transform, DTFT, DFT
- Ideal Sampling and Reconstruction
- Practical ADC/DAC Sampling and reconstruction

Wireless Communication System

Transmitter

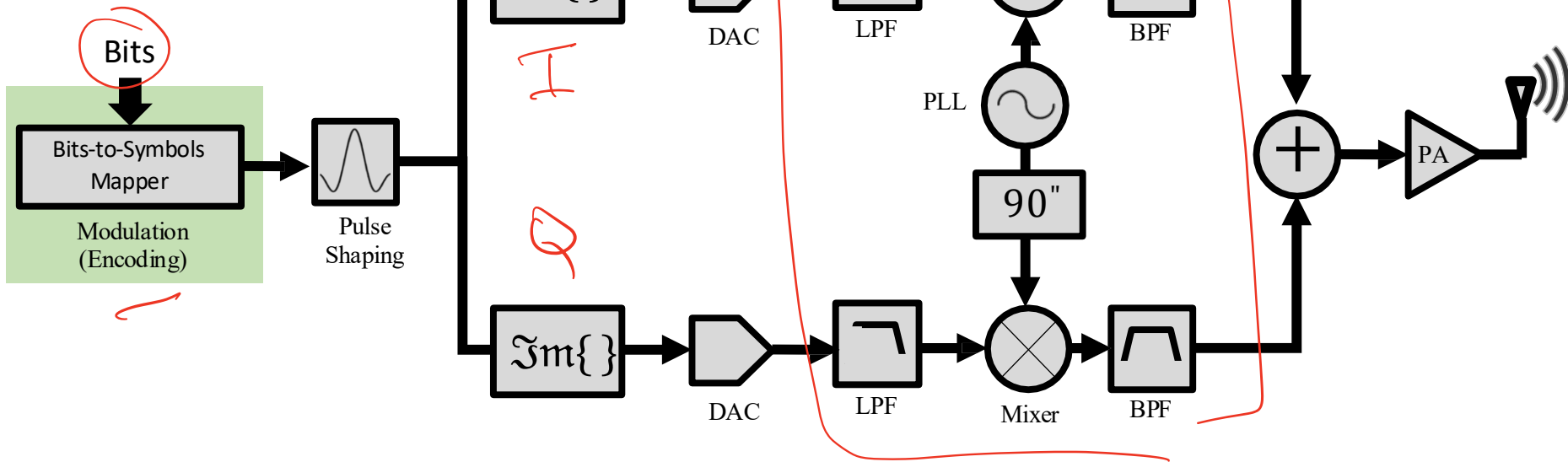


Receiver

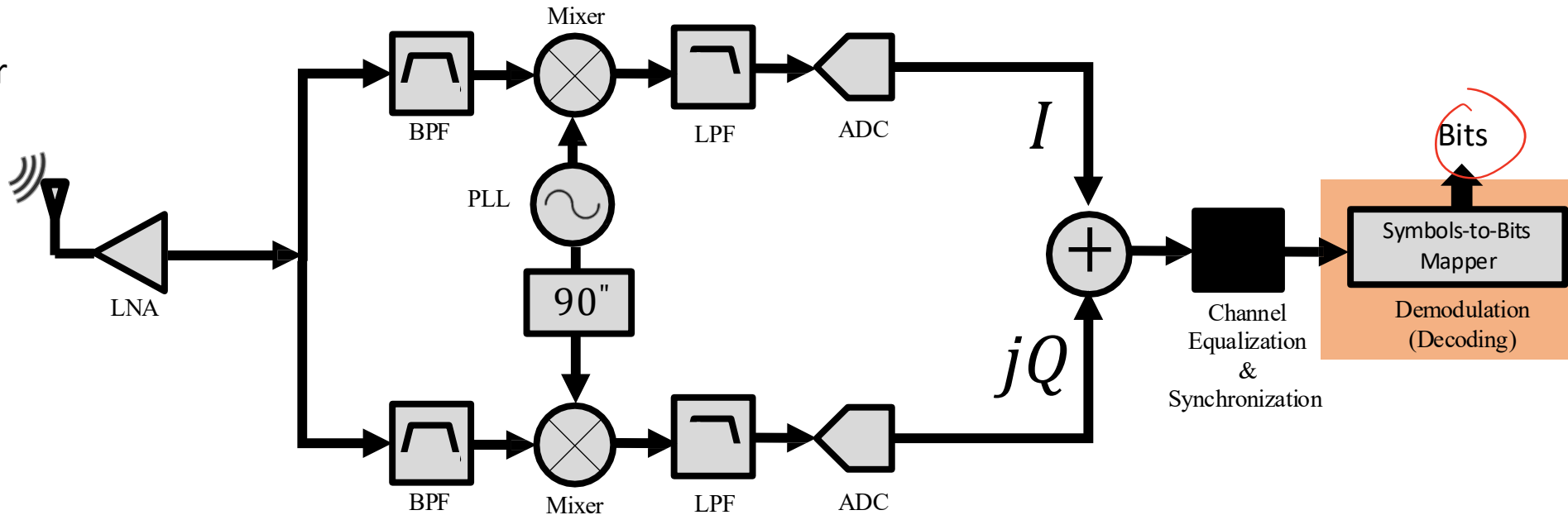


Wireless Communication System

Transmitter



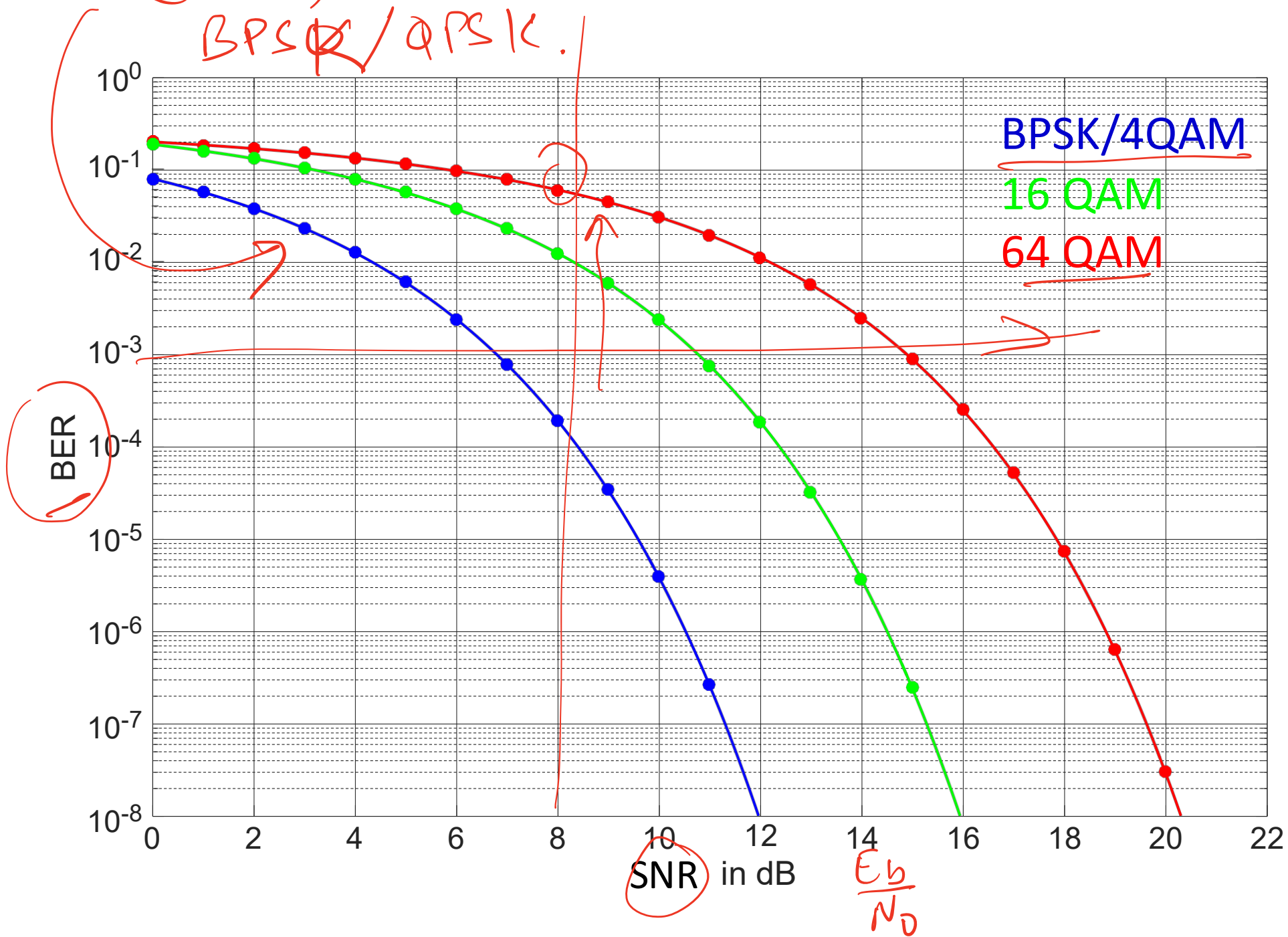
Receiver



$$Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

BPSK/QPSK.

Bit Error Rate (BER)



BPS K

Bit Error Rate (BER)

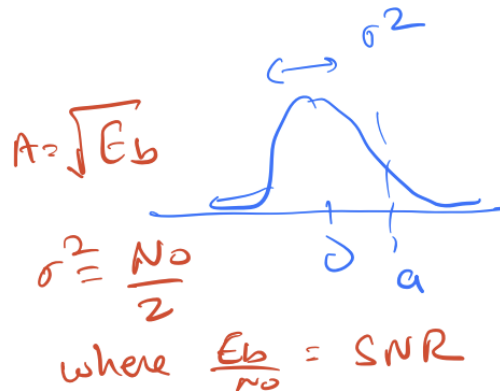
$$\begin{aligned}
 P(\text{bit error}) &= P(y_i > 0 | H_0) \text{ or } P(y_i < 0 | H_1) \\
 &= \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2} \frac{(y_i - -A)^2}{\sigma^2}} = P(y_i | -A)
 \end{aligned}$$

$$\begin{aligned}
 P(-A + v > 0) \\
 \therefore P(v > A) &= P\left(\frac{v}{\sqrt{\sigma^2}} > \frac{A}{\sqrt{\sigma^2}}\right)
 \end{aligned}$$

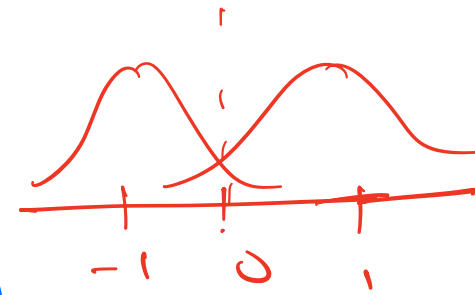
$$\frac{v}{\sqrt{\sigma^2}} \sim N(0, 1)$$

$$\begin{aligned}
 Q \text{ function} &= P(\text{st-normal dist} > a) \\
 Q(a)
 \end{aligned}$$

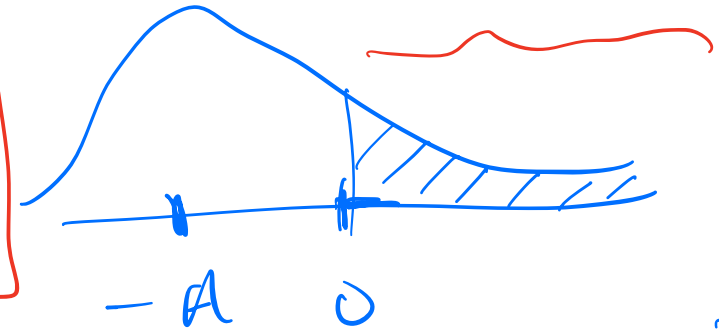
$$\begin{aligned}
 P(v > A) &= Q\left(\frac{A}{\sqrt{\sigma^2}}\right) \\
 &= Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \text{ where } \frac{E_b}{N_0} = \text{SNR}
 \end{aligned}$$



$$\begin{aligned}
 P_e(1|-1) \\
 P_e(-1|1)
 \end{aligned}$$

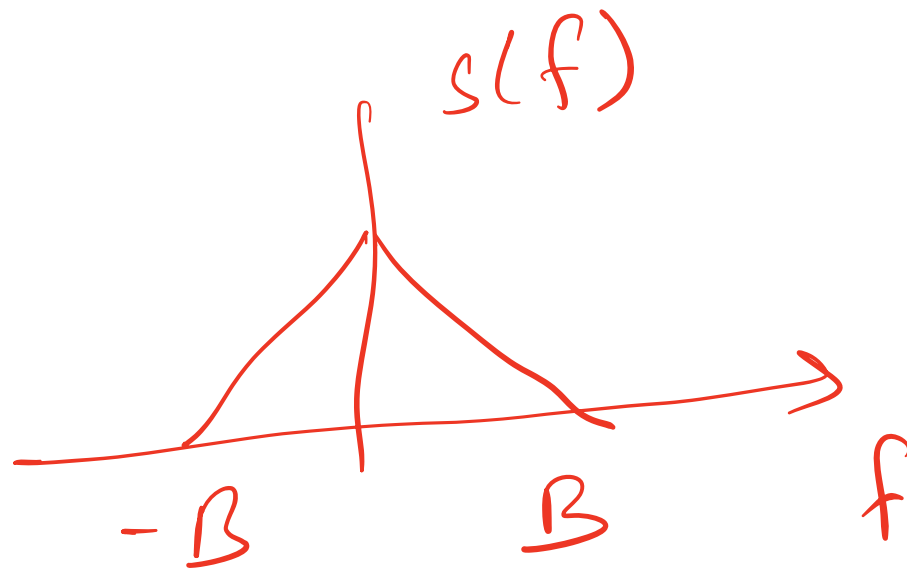
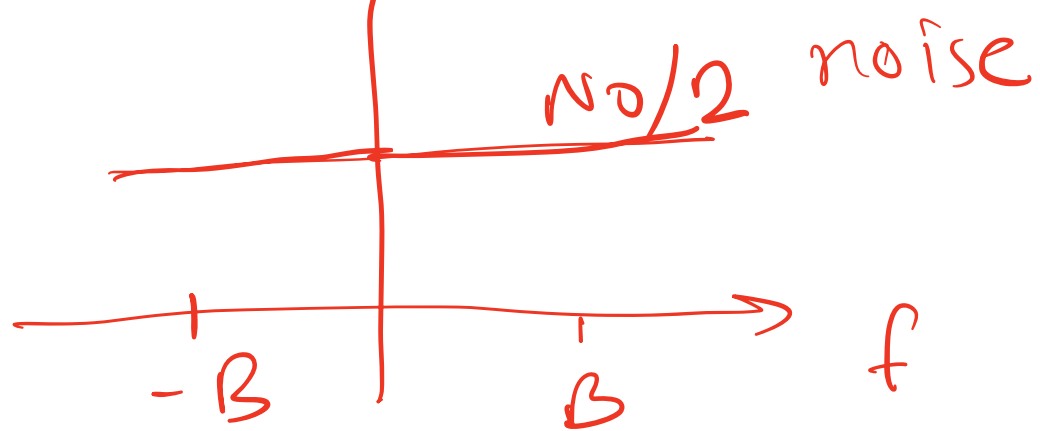


→
No
closed
form



$$\begin{aligned}
 Q(a) &= P(y > a) \\
 y &\sim N(0, 1)
 \end{aligned}$$

$$\begin{aligned}
 P(v > A) &= P\left(\frac{v}{\sqrt{\sigma^2}} > \frac{A}{\sqrt{\sigma^2}}\right) \\
 &= Q\left(\frac{A}{\sqrt{\sigma^2}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)
 \end{aligned}$$



$$\int_{-B}^B () = N_0$$

$$Q \sqrt{\frac{2E_b}{N_0}}$$

$$= Q \sqrt{2 \cancel{E_b} NR}$$

Symbol Error Rate (SER)

16 QAM = $2^4 = 4$ ~~bits~~ bits/symbol

bit error = P_b for each bit

what is symbol error rate?

SER

P_s = Prob that ~~any~~ one bit in error
at least

1 - no bits in error

= 1 - bit 1 no error $(1 - P_b)$
x bit 2 no error $(1 - P_b)$
:

= $1 - (1 - P_b)^k$ k^{th} order mod.

QPSK

$k = 2$

$$P_s = 1 - (1 - P_b)^2$$

$$\approx 2P_b$$

$$P_b = \frac{P_s}{2}$$

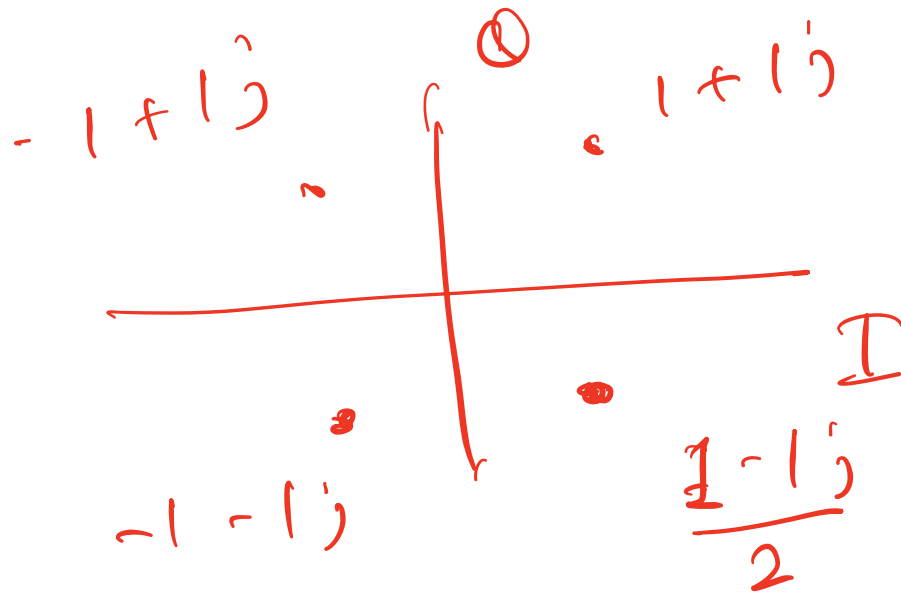
BER & SER for QPSK

2 bits/symbol

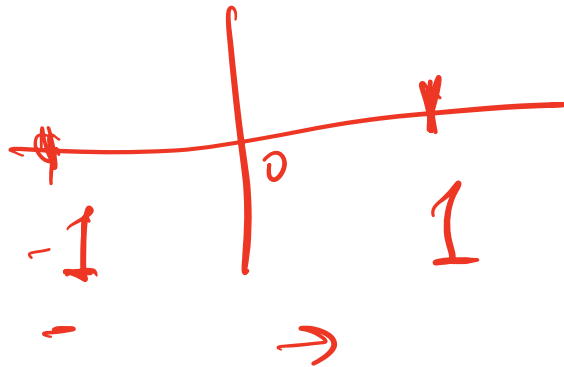
$$E_s = (1)^2 + (1)^2$$

$$= 2$$

$$E_b = 1$$

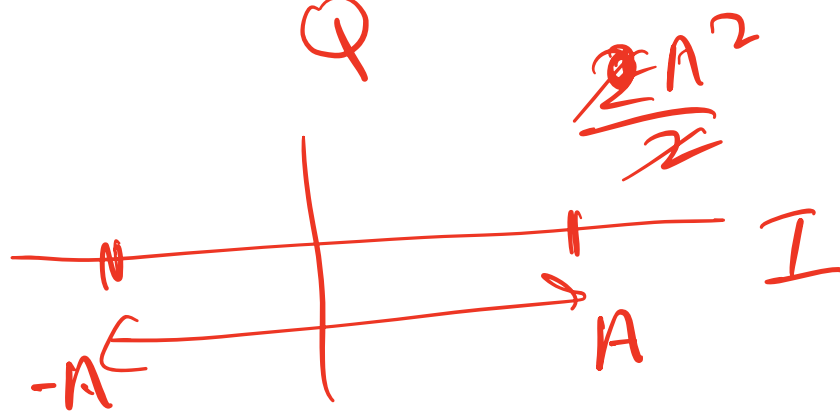


BPSK



$$(1)^2 + (-1)^2 = 2$$

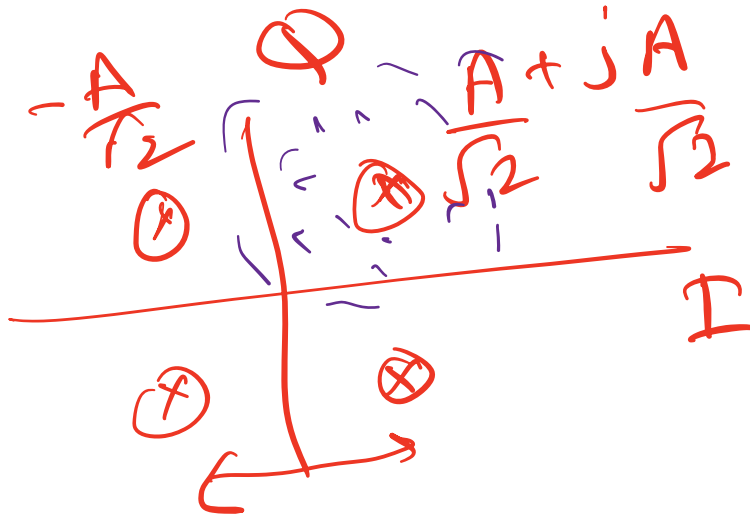
BPSK



Prob error

$$BER = SER \text{ for BPSK}$$

QPSK



SER_{QPSK} is Higher than SER_{BPSK}

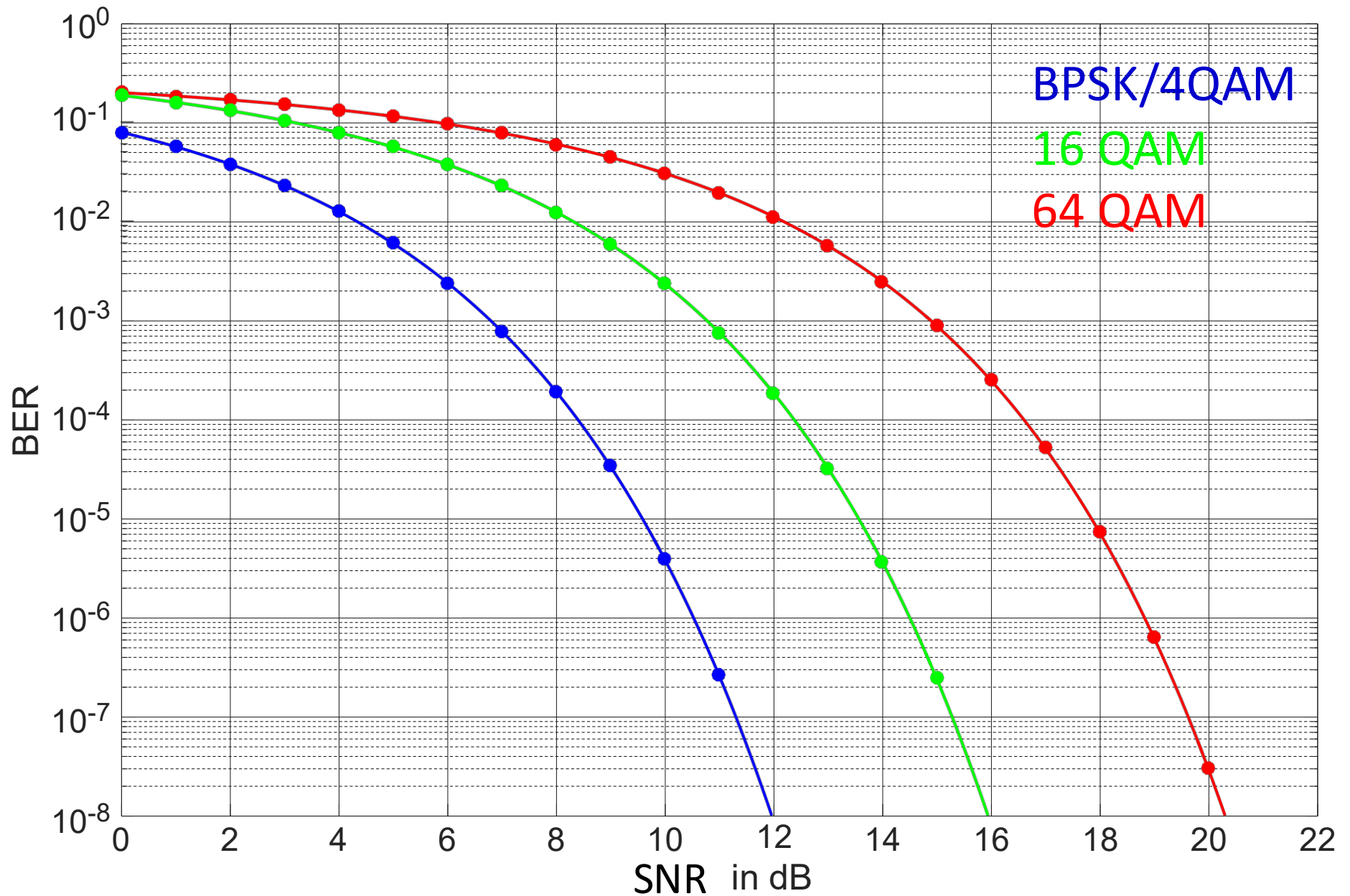
BER

$$\text{Energy} = \left(\frac{A}{\sqrt{2}}\right)^2 + \left(\frac{A}{\sqrt{2}}\right)^2 = A^2$$

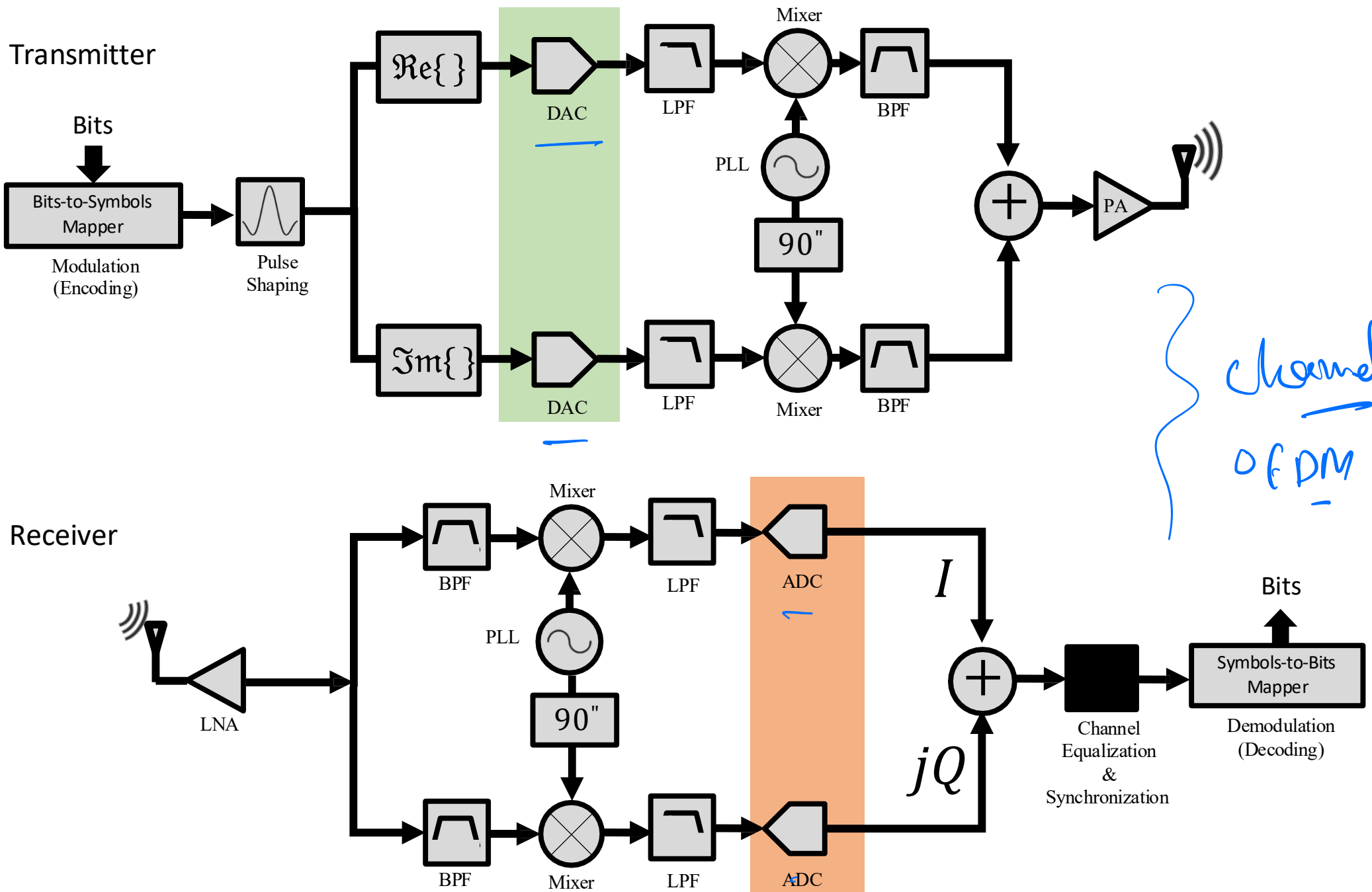
$$\begin{aligned} z &\sim N(0, \sigma^2) \\ &\sim N\left(0, \frac{N_0}{2}\right) \end{aligned}$$

$$\begin{aligned} BER_{QPSK} &\approx BER_{BPSK} \end{aligned}$$

Bit Error Rate (BER)



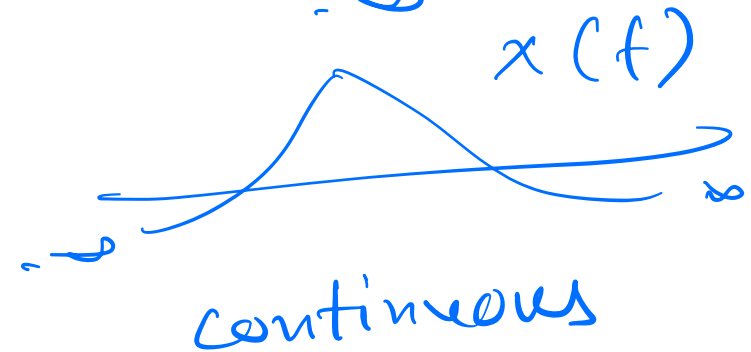
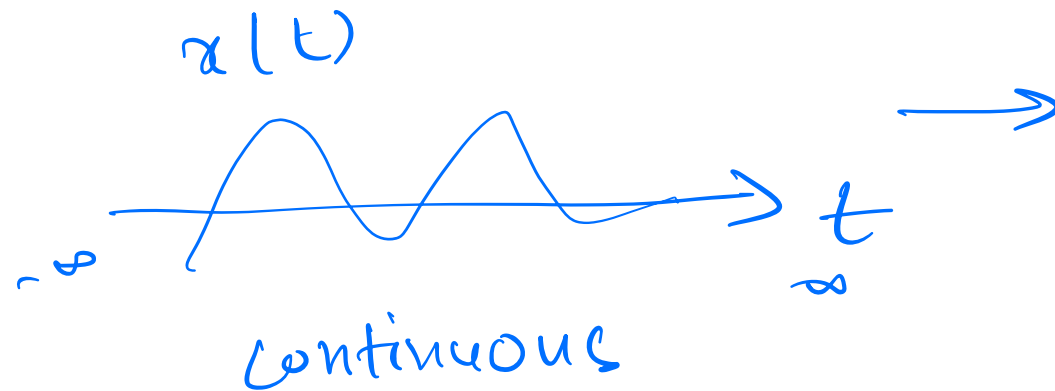
Wireless Communication System



Review: DTFT, DFT

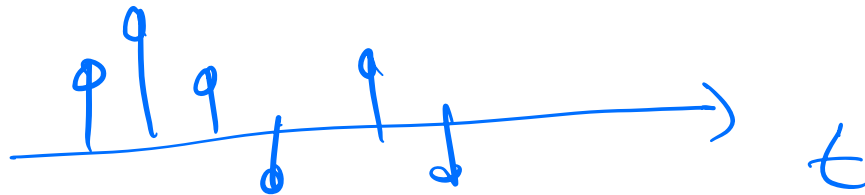
Fourier transform
 $x(t)$

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$



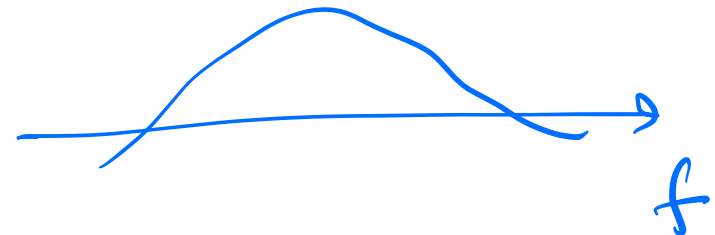
DTFT

$$t = nT_s$$



discrete

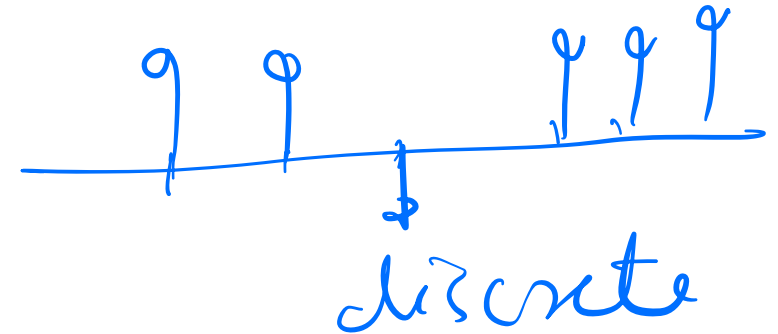
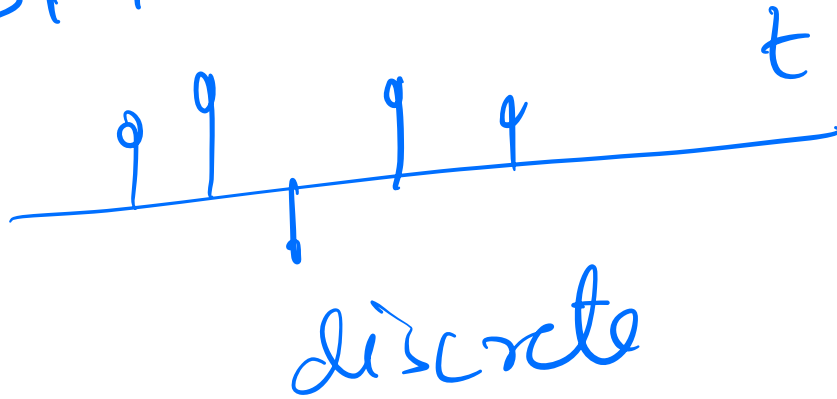
$$X(f) = \sum x(nT_s) e^{-j2\pi f nT_s}$$



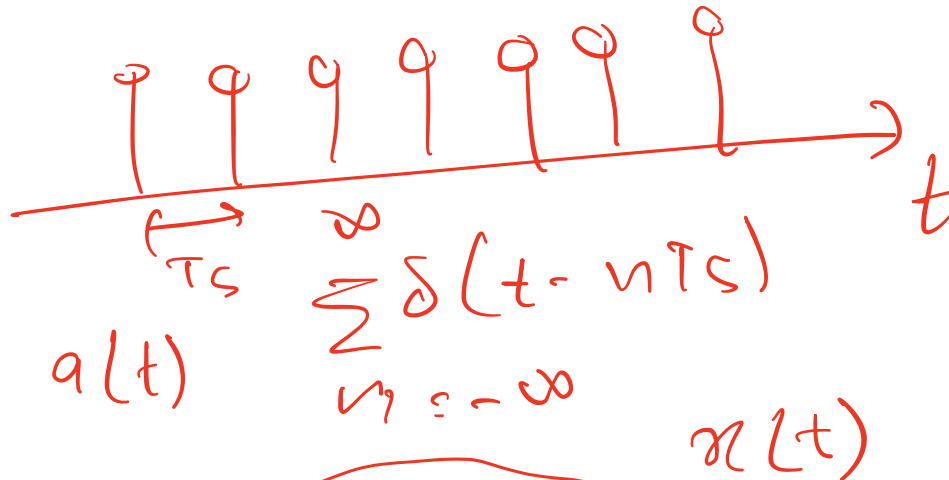
continuous

Review: DTFT, DFT

DFT



Pulse train

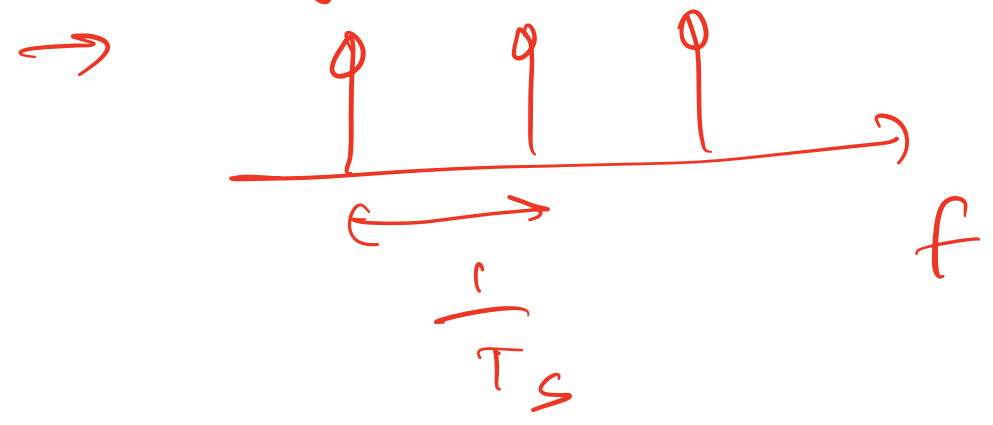
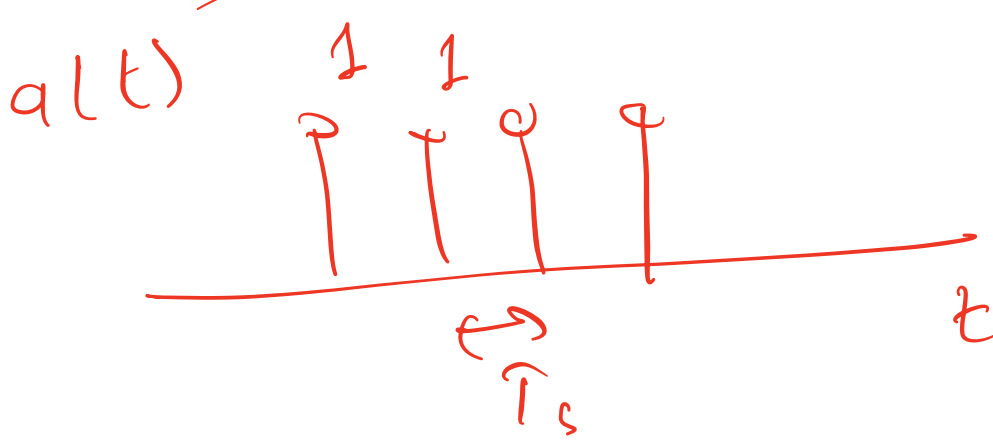
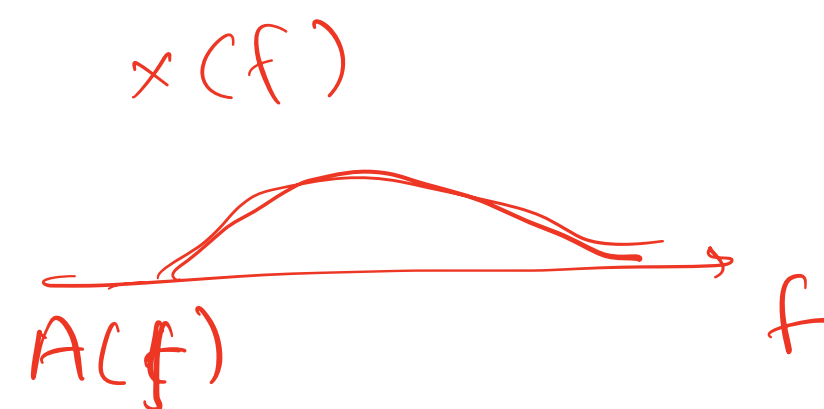
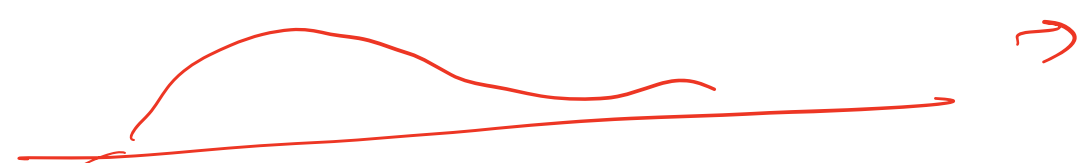


$$x_\delta(t) = a(t) \times x(t)$$

$$= x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$\times \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

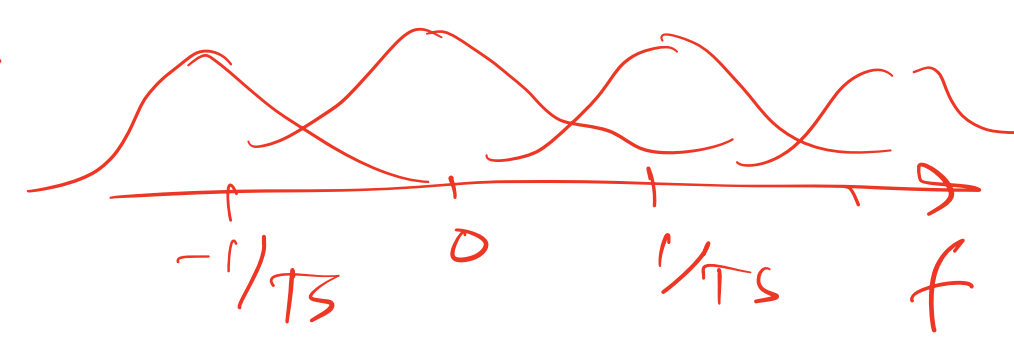
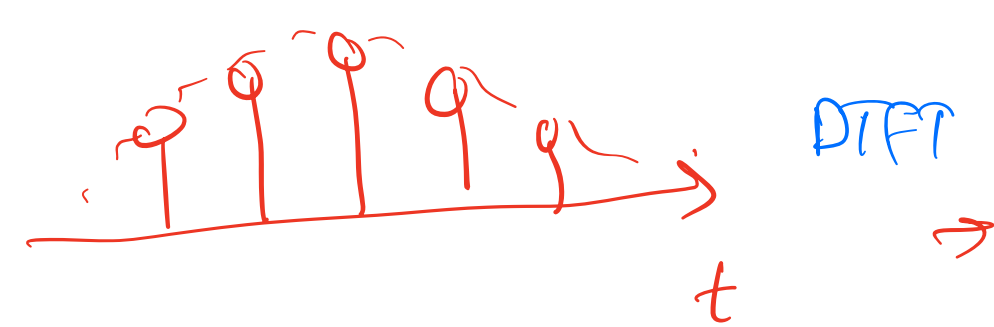
$x(t)$ $x(t)$ DTFT



$$x_\delta(t) = x(t)a(t)$$

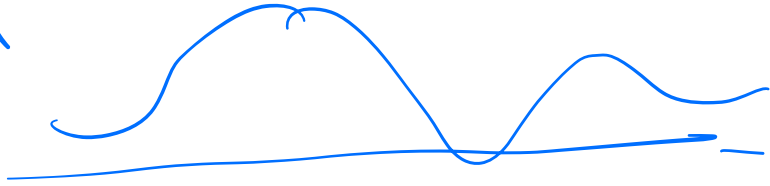
$$X_\delta(f) = x(f)$$

Periodic $\otimes A(f)$



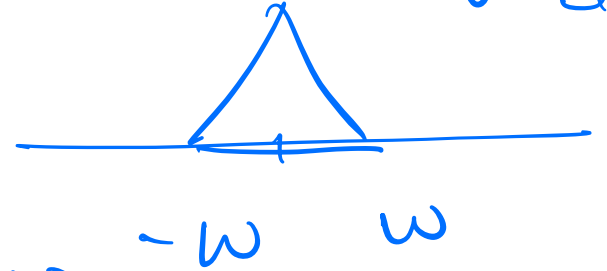
Aliasing

$x(t)$
real

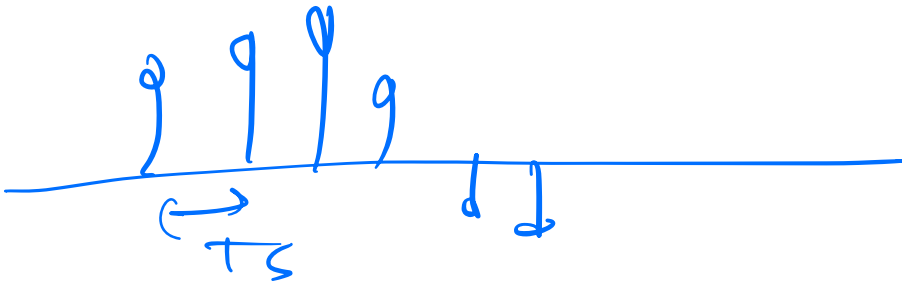


$x(f)$

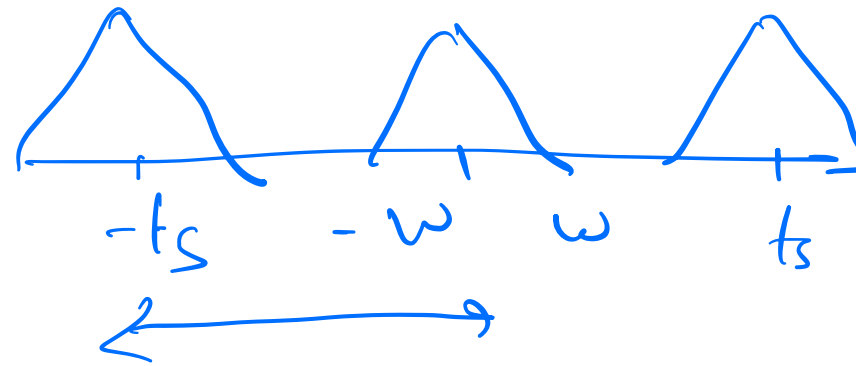
symmetric



$x_s(t)$



$x_s(f)$



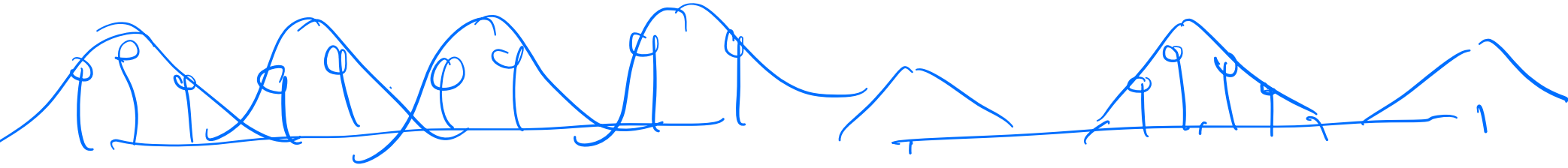
Nyquist

$$\underline{f_s > 2w}$$

$$f_s = \frac{1}{T_s}$$

DFT

$$x_s(f) \times \delta(f)$$



discrete
& periodic

discrete &
Periodic

represented by
samples in
one period

Ideal Sampling

Ideal sampling

Sampling using "delta" functions

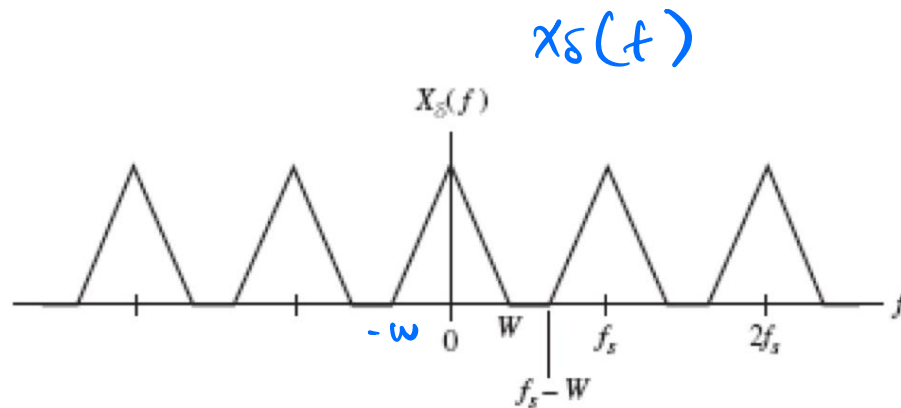
$$x_{\delta}(t) == \sum_{k=-\infty}^{\infty} x(kT_s) \delta(t - kT_s)$$

$$= x(t) \sum_{k=-\infty}^{\infty} \delta(t - kT_s)$$

Spectrum

$$X_{\delta}(f) = f_s \sum_{n=-\infty}^{\infty} X(f - nf_s)$$

$f_s = \frac{1}{T_s}$



ADC

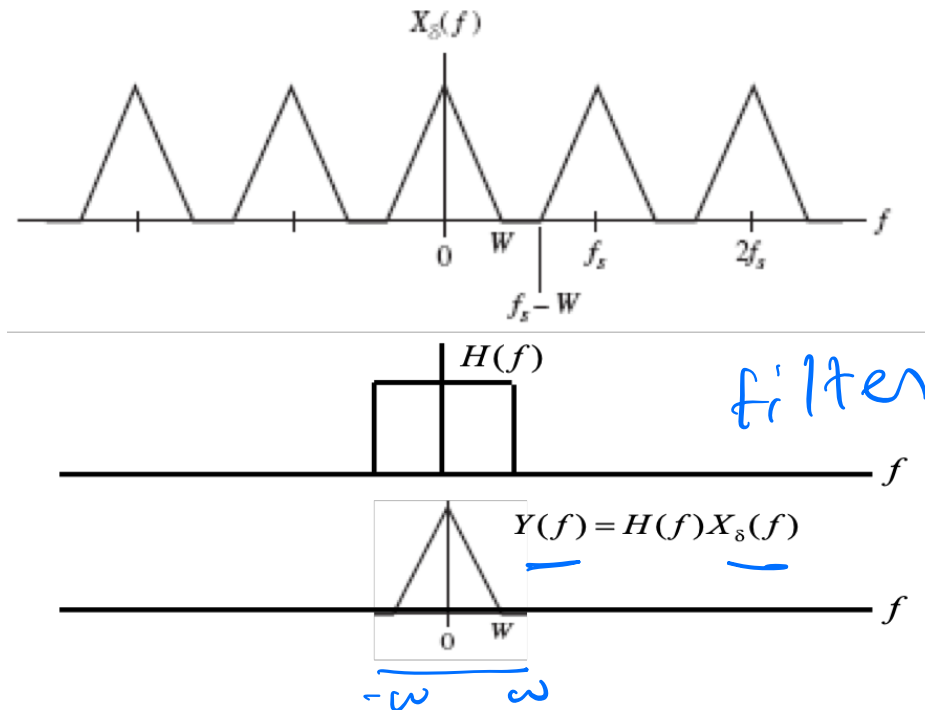
Nyquist

$$f_s \geq 2W$$

Ideal reconstruction (1/3)

Use an LPF

With cutoff between W and $f_s - W$

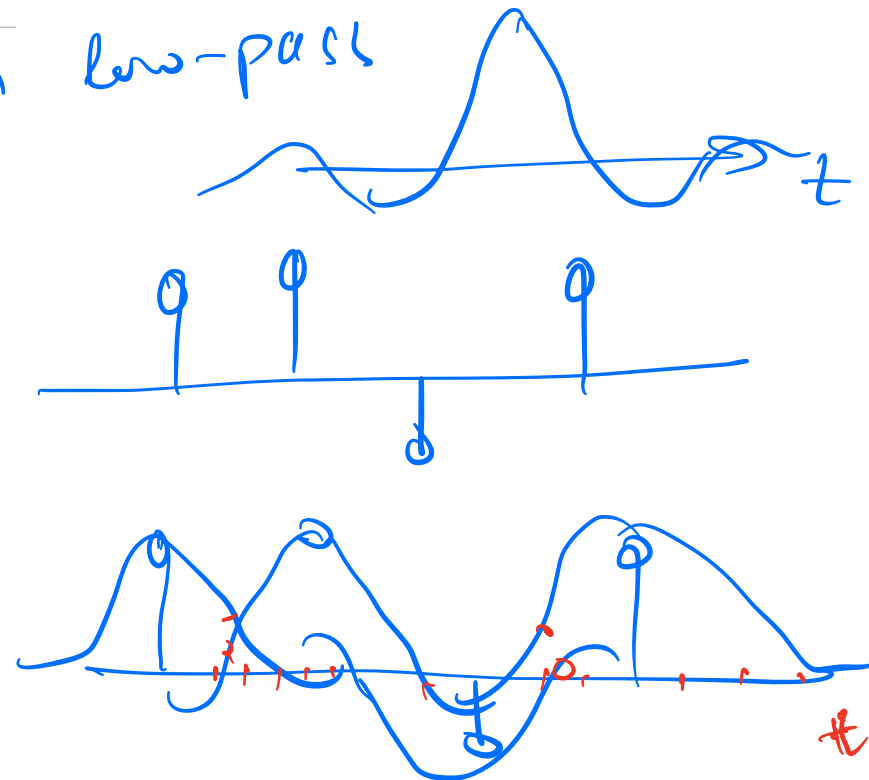


© 2010 The McGraw-Hill Companies

DAC

$$\underline{x_s(t)} \otimes h(t)$$

$h(t)$

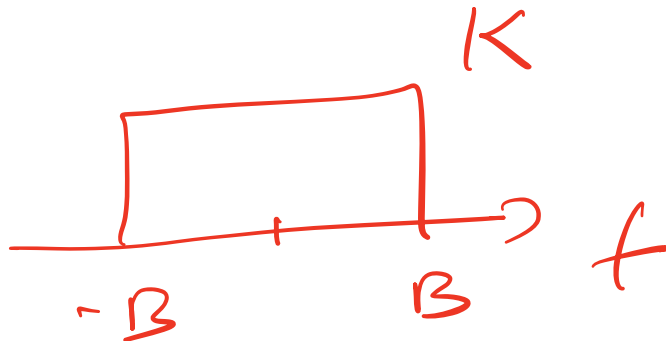


Ideal reconstruction (2/3)

Ideal LPF

$$H(f) = K \Pi\left(\frac{f}{2B}\right) e^{-j2\pi f t_d}$$

Reconstructed signal is like original signal, but delayed by t_d , gain K , bandwidth B



$$B \geq \omega$$

Ideal reconstruction (3/3)

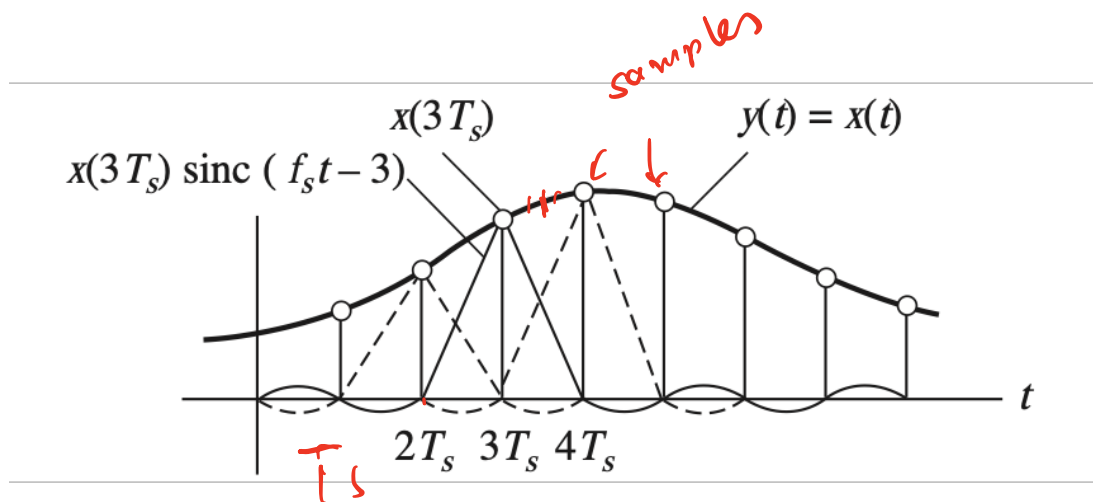
Reconstruction by interpolation (time domain)

Reconstruct from samples, but with ideal interpolation

$$h(t) = \mathcal{F}^{-1}[H(f)] = 2BK \text{sinc} 2B(t - t_d)$$

$$\Rightarrow y(t) = h(t) * x(kT_s) = 2BK \sum_{k=-\infty}^{\infty} x(kT_s) \text{sinc} 2B(t - t_d - kT_s)$$

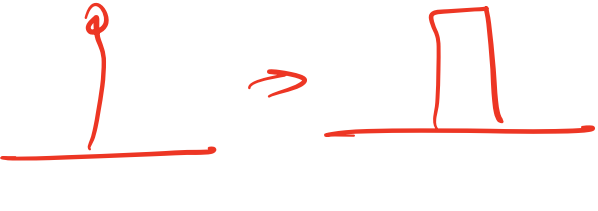
(Unimplementable in practice)



Practical sampling & reconstruction

Real samplers

- Have finite duration pulses (not delta functions)



Real reconstruction filters

- Not ideal (or distortion-less filters)
- Must be causal (in most cases)

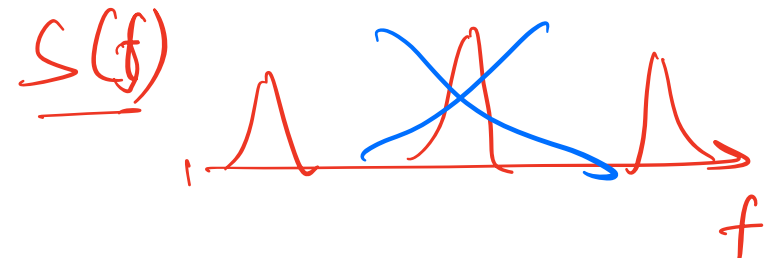
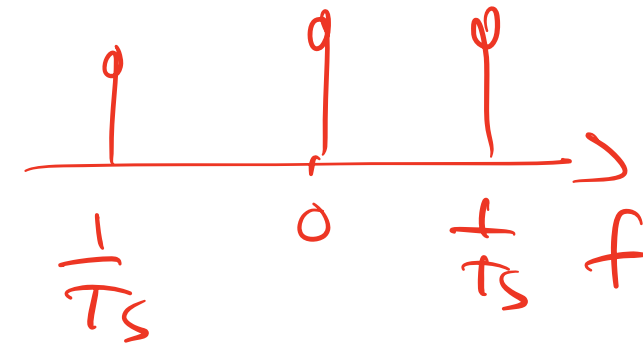
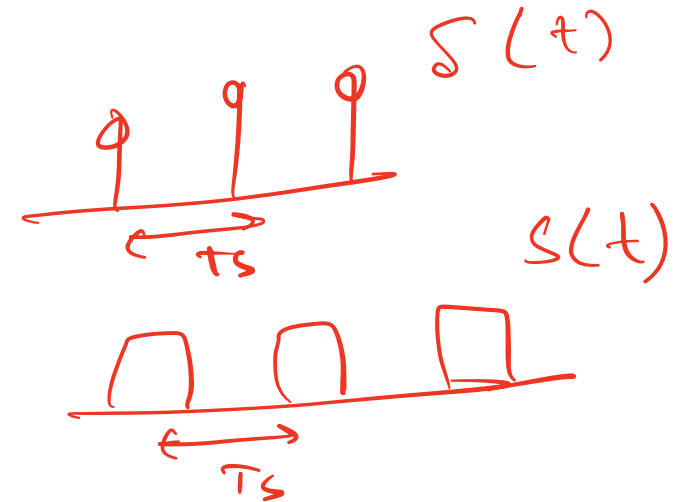
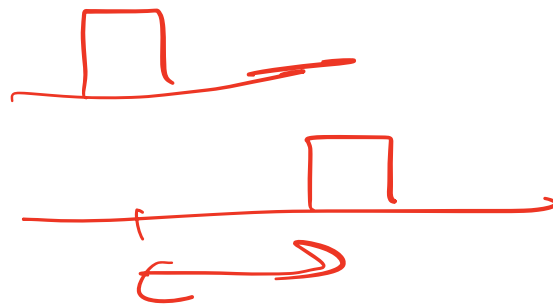
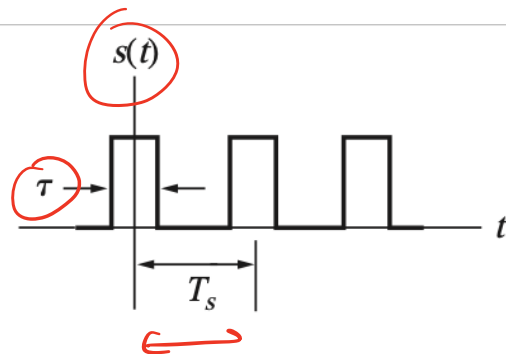
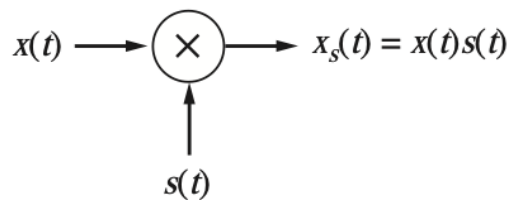
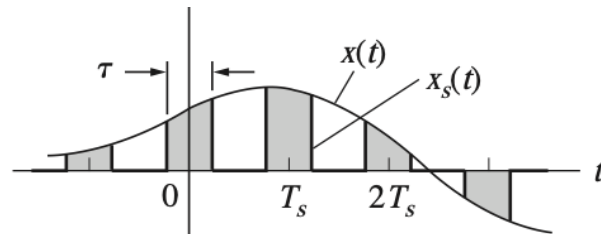
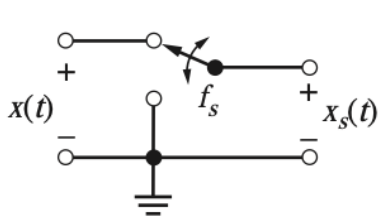
Real signals

- Are time-limited, and therefore cannot be band-limited
- $\Rightarrow$ Signal *distortion* and *aliasing* cannot be entirely avoided when using sampling and reconstruction

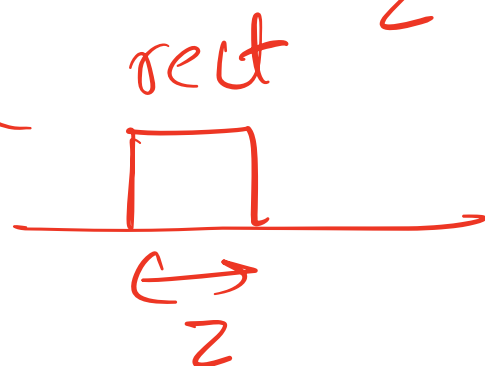
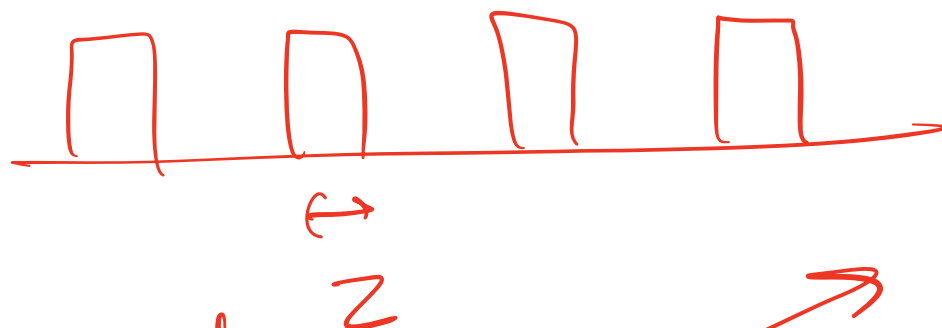
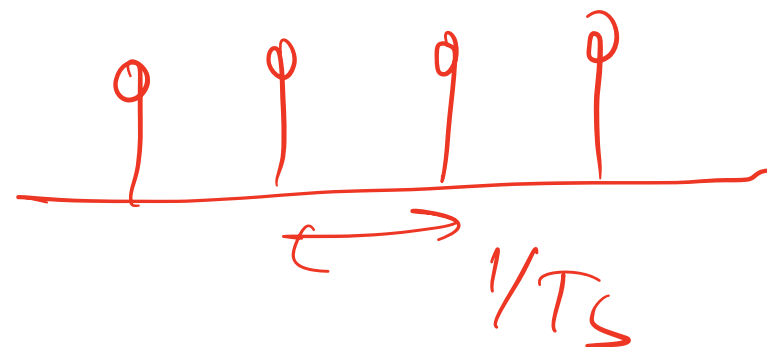
Non-ideal sampling (chopper sampling)

Pulse sampling or chopper sampling

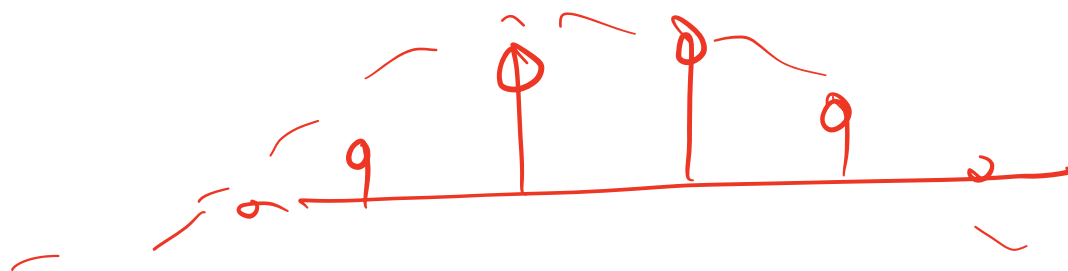
$x_s(t)$: A sampled version of $x(t)$

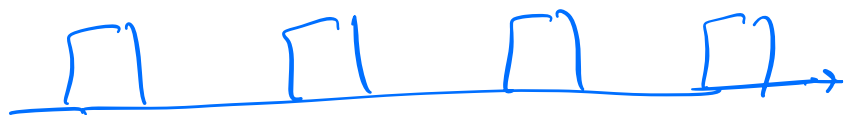


(A)



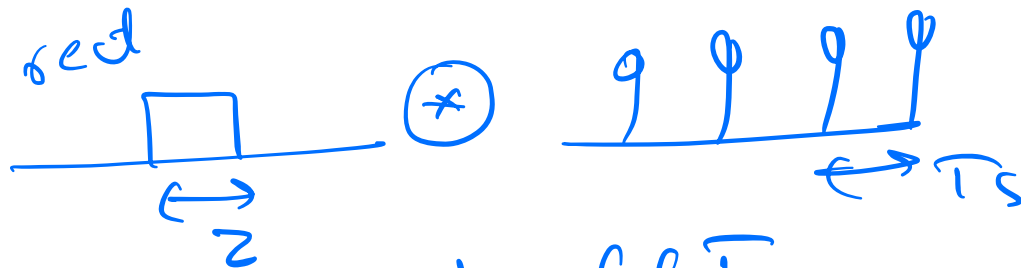
fft





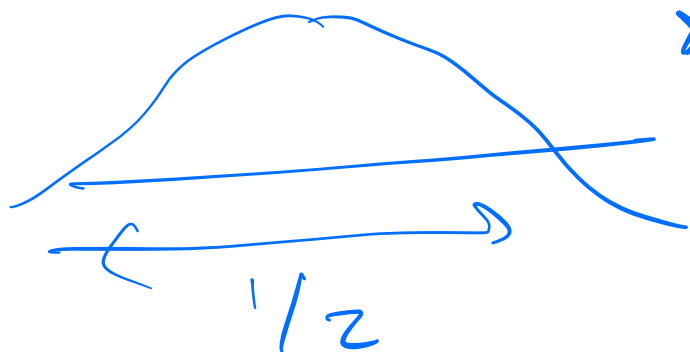
$$z \leq T_s$$

= sed

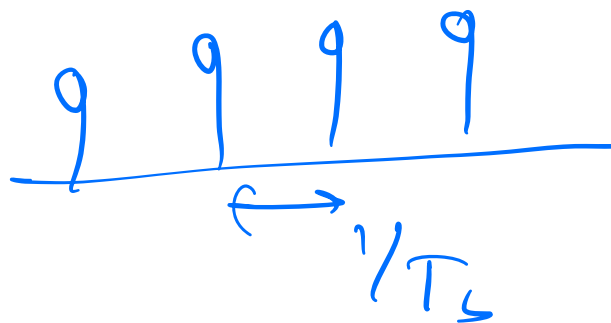


$\downarrow$ FFT

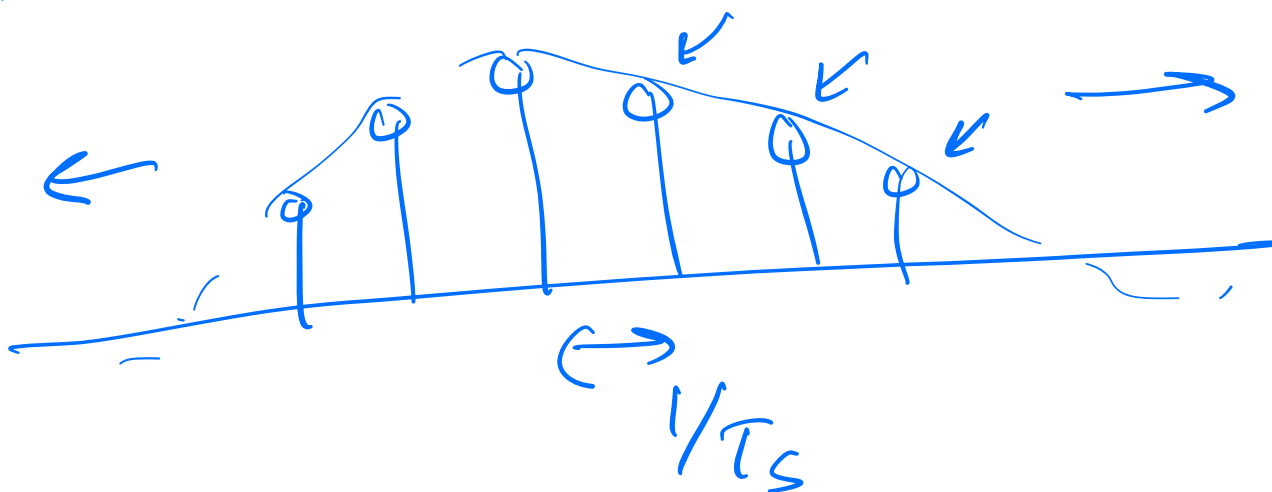
sinc



$\times$



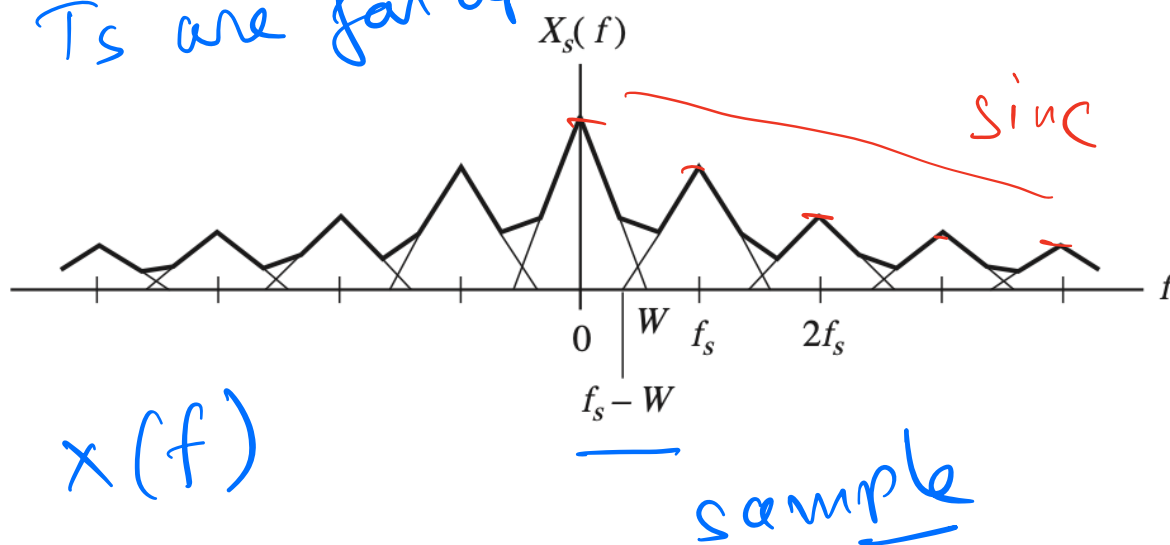
$$z \rightarrow 0$$



Aliasing

Under-sampled message

T_s are far apart

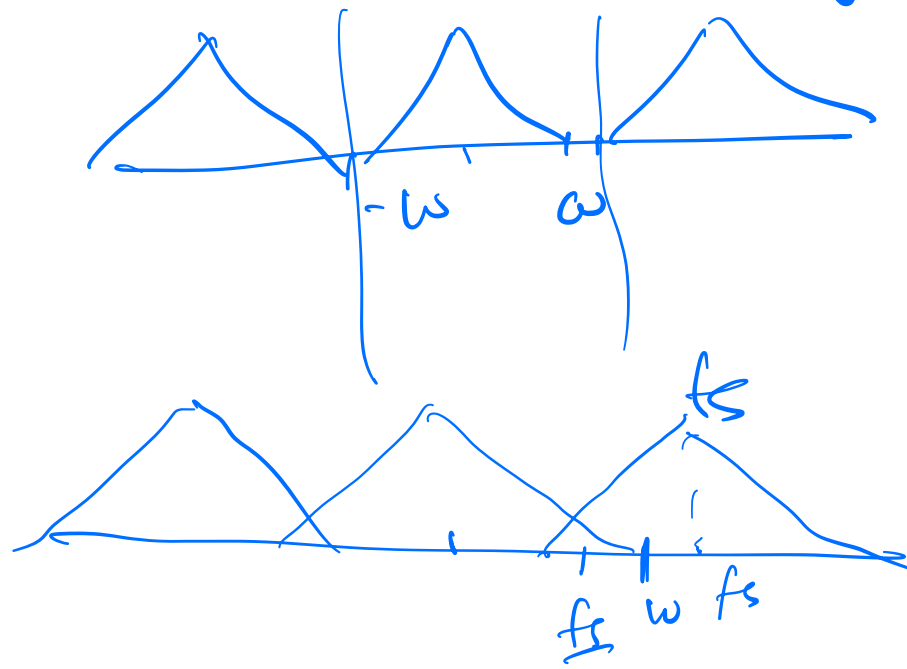
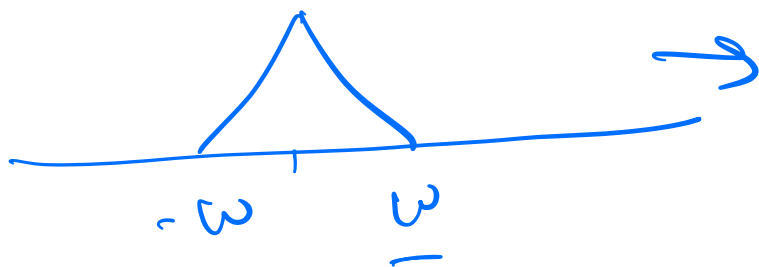


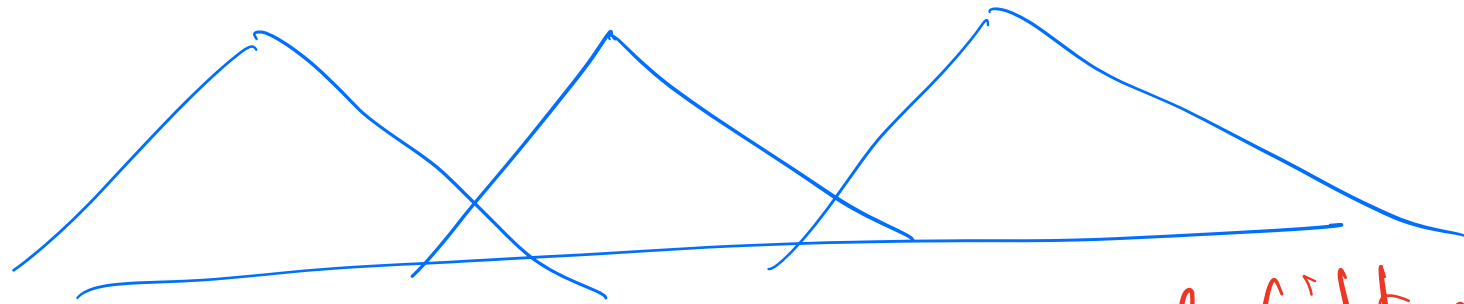
$$\underline{f_s} \geq 2W$$

No aliasing

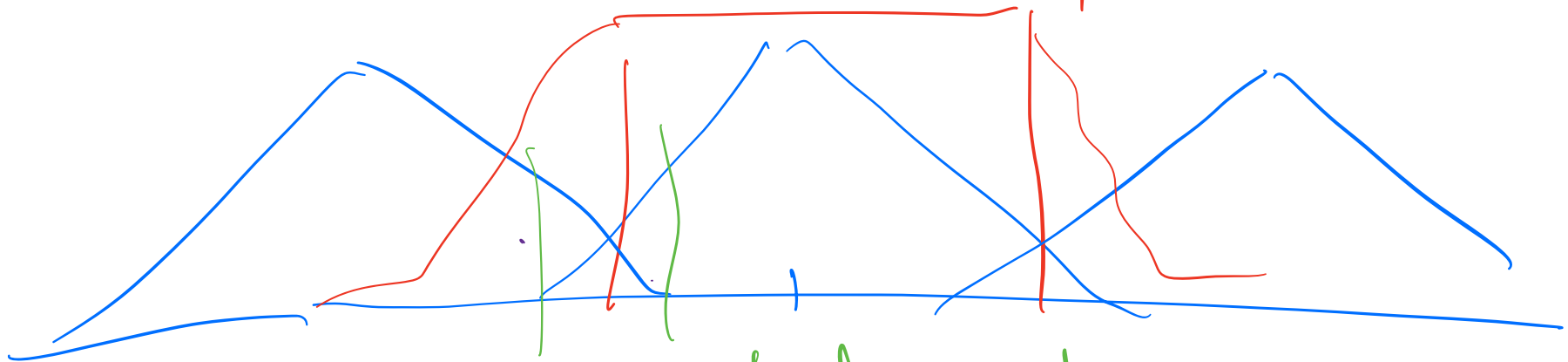
$$f_s < 2W$$

aliasing

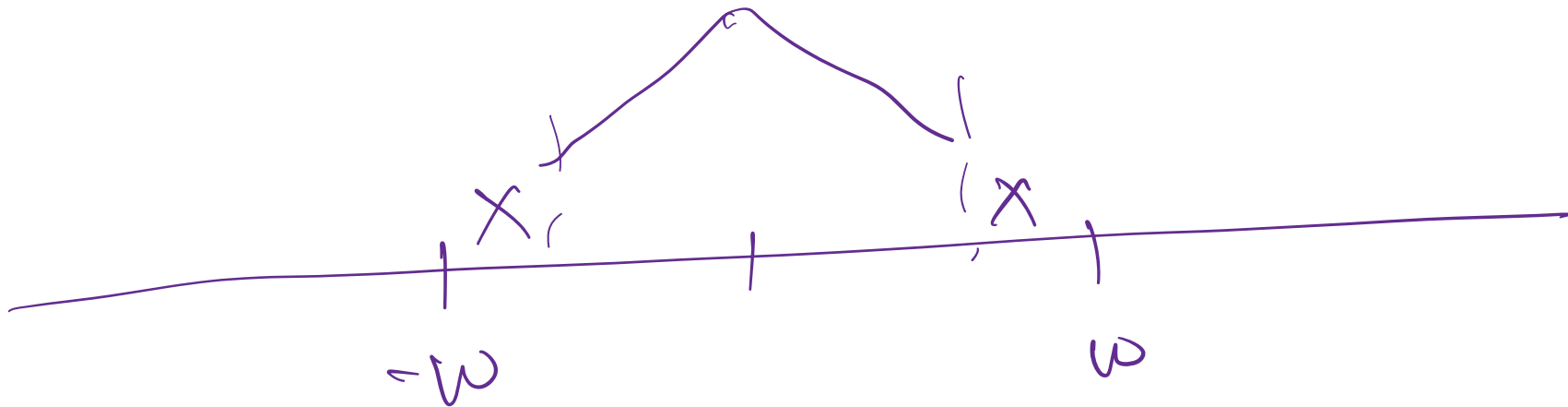


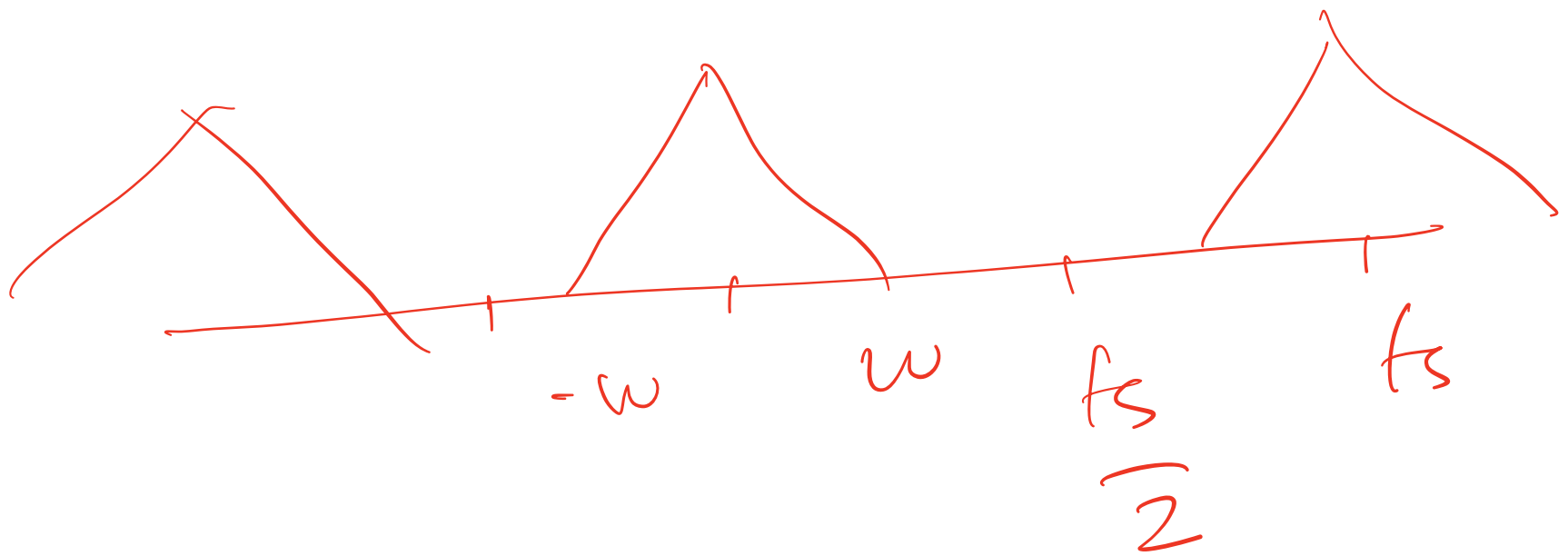


ideal filter



distorted spectrum.

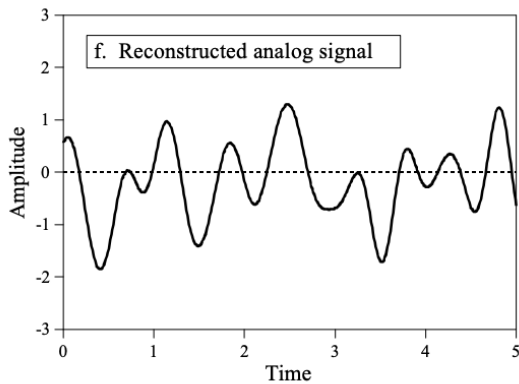
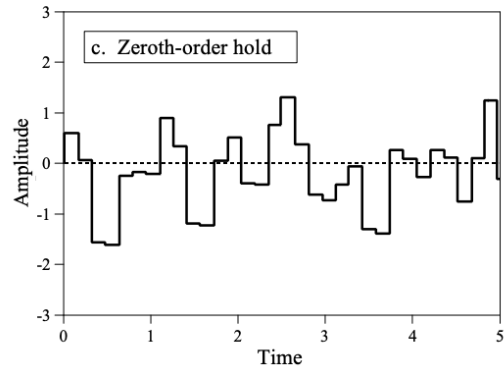
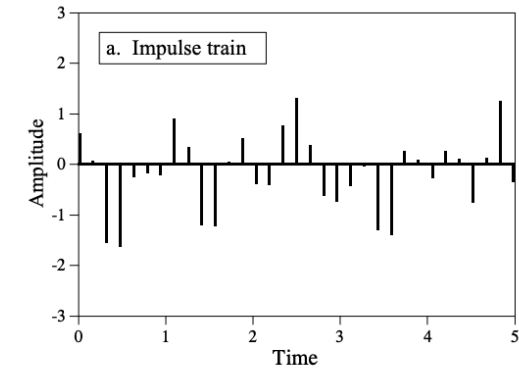




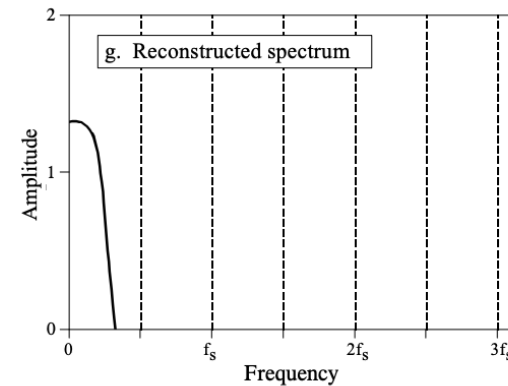
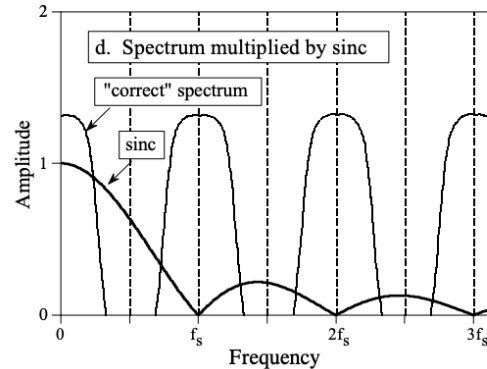
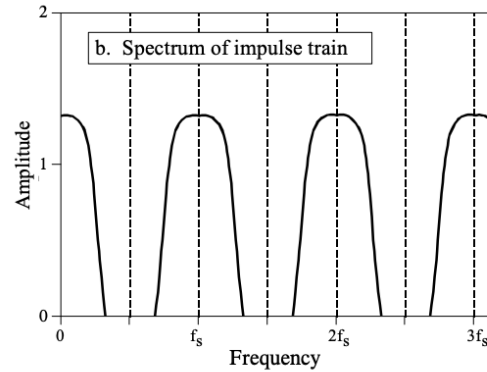
$$f_s \gg 2w$$

DAC Design and reconstruction

Time Domain

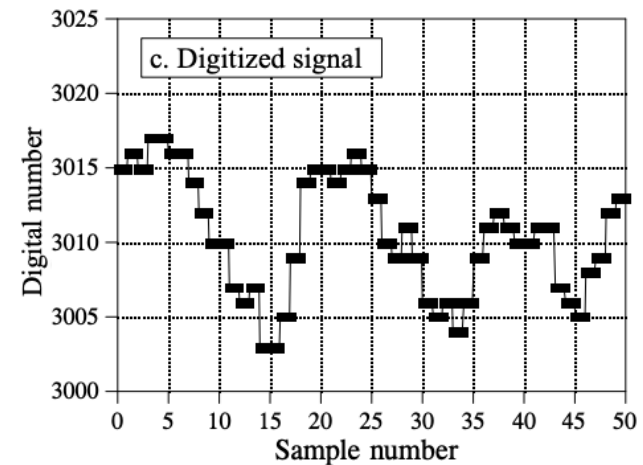
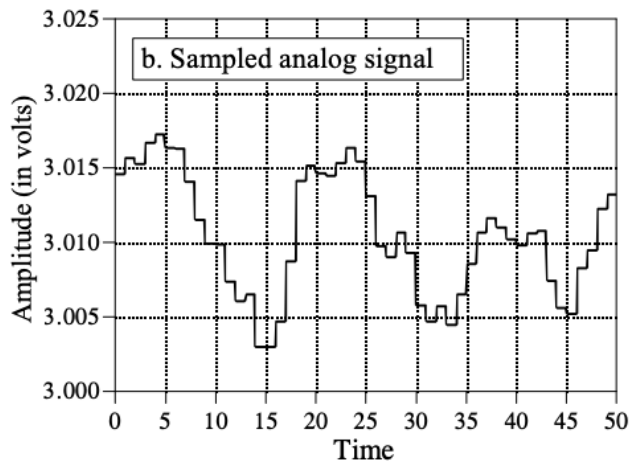
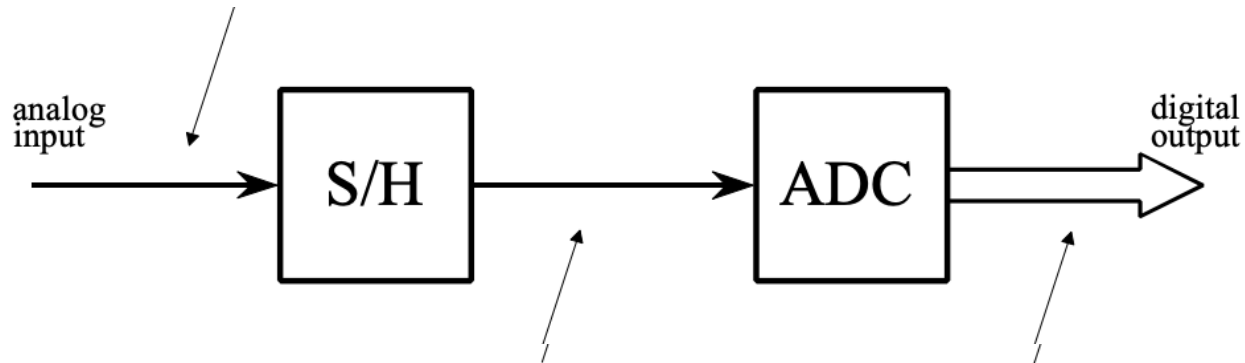
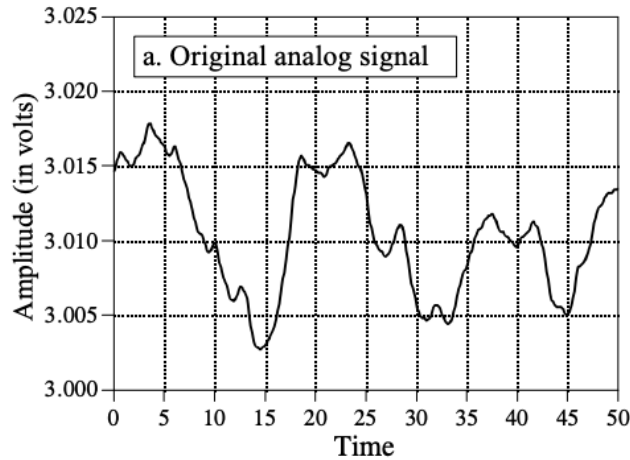


Frequency Domain



DAC Design and reconstruction

ADC Design with Quantization



ADC Design with Quantization